MAPLE & MAIN · DEMAND PLANNING

A forecast is only worth the ordering decision it improves

Weekly demand forecast at four planning levels, tested the way a buying team would use it, and priced in markdown and lost margin.

Synthetic data. No client, employer or proprietary information.

EXECUTIVE SUMMARY

Forecast error costs 7.4% of store revenue. Most of it is unavoidable; $2.5M a year is not.

$2.2M

a year saved by forecasting properly rather than repeating last year, measured as markdown plus lost margin on store-level ordering.

$256k

a year more from ordering to the right service level instead of to the forecast itself. The forecast is the middle of a range, not the plan.

2.2%

error at company level and 23.1% at store and category level. Accuracy is a property of the level you plan at, not of the model.

Recommendation: run a different method at each level rather than one model for everything, publish a range rather than a number, and set order quantities from the 63% service level the margin and markdown rates imply.

APPROACH

Tested the way the business would use it

Ten forecasts made four weeks apart, each judged up to thirteen weeks ahead: the lead time a buying team works to. 2025-11-03 to 2026-10-05.

1. Four planning levels

Company, category, store, and store by category: 549 weekly series in all.

SQL · 156 weeks

2. Benchmarks first

Repeat last week, repeat last year, and repeat last year adjusted for the current level. A model has to beat these.

Three benchmarks

3. Only what is known

Trend, seasonality, the trading calendar and the media plan. No input that would not be available on the day.

No peeking

4. Priced, not just scored

Error converted to markdown and lost margin, and the ordering rule optimised against it.

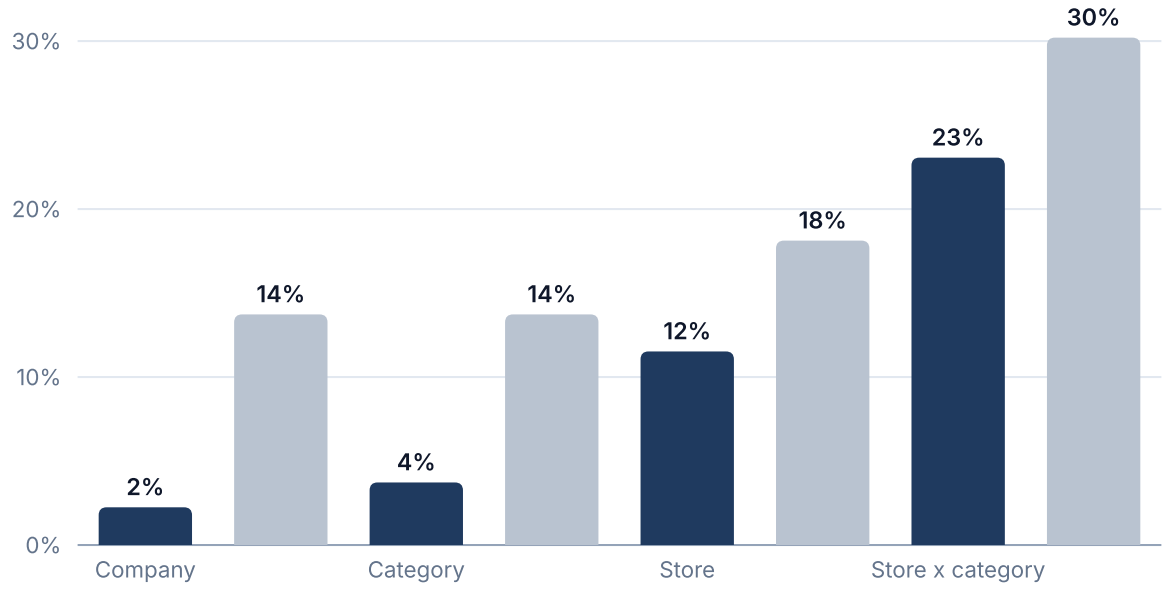
Newsvendor economics

DATA USED	CHECKS BEFORE MODELLING	HOW RESULTS WERE VALIDATED
Every order line for three years, aggregated to Monday weeks, with the media plan and the trading calendar	Category revenue reconciles to the company total to the dollar; no missing weeks; no lines outside the window	Rolling-origin backtest on ten origins; intervals fitted on the first five origins and tested on the last five

WHERE ACCURACY GOES

Error roughly doubles at every step down the hierarchy

Forecast error by planning level, best method at each



Weighted absolute percentage error across the backtest. Dark bars: best method. Light bars: repeating the same week last year.

LEVEL	SERIES	ERROR	BENCHMARK
Company	1	2.2%	13.7%
Category	8	3.7%	13.7%
Store	60	11.5%	18.1%
Store x category	480	23.1%	30.2%

Aggregation hides noise. A company forecast is accurate because individual weeks cancel out, not because the model understands the business better. Quote accuracy with the level attached, or it means nothing.

METHOD SELECTION

No single method wins everywhere, and the most complex one wins nowhere

METHOD	COMPANY	CATEGORY	STORE	STORE X CATEGORY
Last week repeated	11.8%	20.9%	20.0%	37.7%
Same week last year	13.7%	13.7%	18.1%	30.2%
Same week last year, level adjusted	2.5%	**3.7%**	15.7%	32.0%
Regression: trend and season	3.5%	5.8%	11.7%	**23.1%**
Regression: plus trading calendar	3.1%	5.4%	**11.5%**	23.2%
Regression: plus media plan	3.0%	5.6%	11.8%	23.8%
Last year and regression, averaged	**2.2%**	3.9%	12.5%	25.3%
Boosted model, all inputs	7.0%	9.0%	14.3%	26.7%

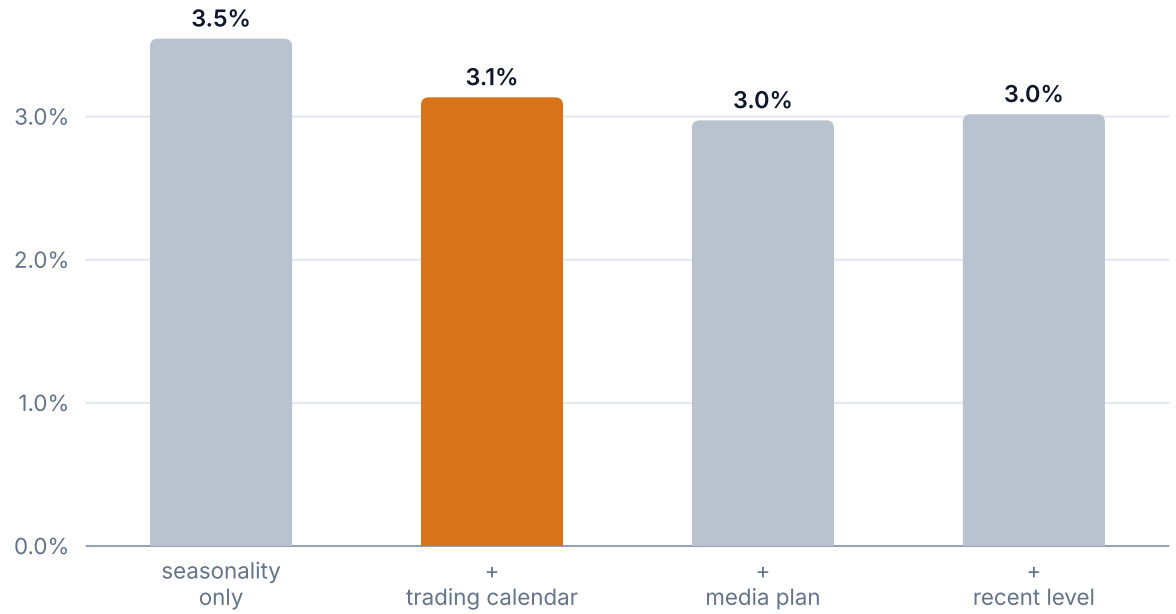
Weighted absolute percentage error. The best figure in each column is highlighted.

- **At company and category level**, last year's shape with the level reset to the last eight weeks is hard to beat. Three years of history is only two full seasonal cycles, and a regression cannot learn the holiday peak better than simply looking at it.
- **At store and store-by-category level**, the regression wins: individual series are too noisy for last year's actual to be a reliable guide.
- **The boosted model is worst at every level.** With 150 weekly observations per series there is nothing for it to learn that the simpler models have not already taken.

WHAT IS WORTH KNOWING

The trading calendar is worth more than the media plan

Error as each kind of knowledge is added, company level



Same model and same backtest throughout; only the inputs change.

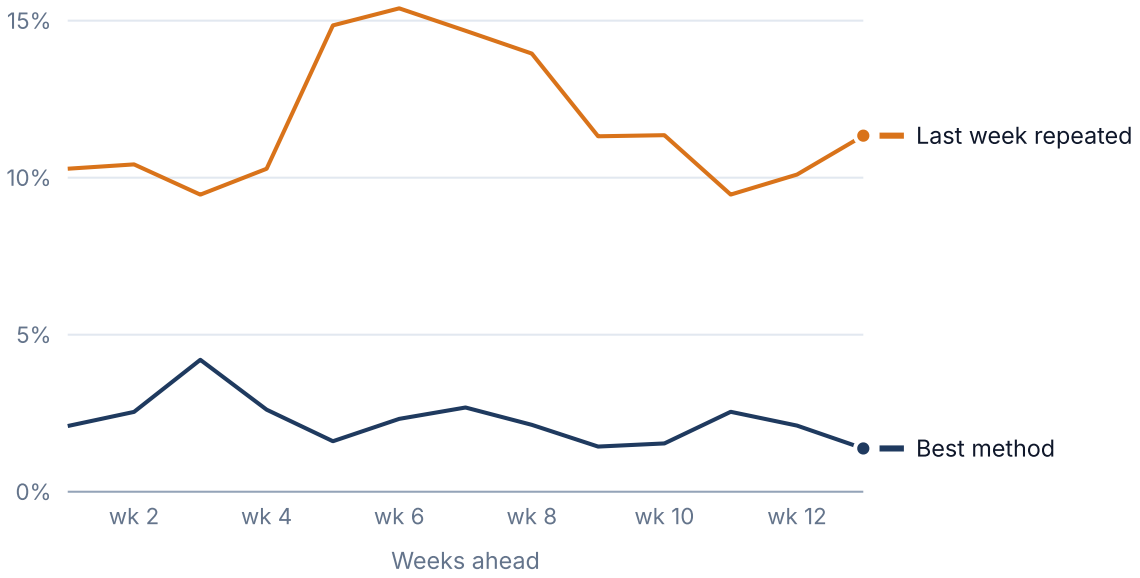
INPUTS	COMPANY	CATEGORY	STORE X CATEGORY
Seasonality only	3.5%	5.8%	23.1%
Plus trading calendar	3.1%	5.4%	23.1%
Plus media plan	3.0%	5.6%	23.8%
Plus recent level	3.0%	5.7%	24.4%

Knowing **when** the flyer drops cuts company-level error by 11.6%. Knowing **how much** is being spent on each channel adds nothing beyond that, and at store level it makes things slightly worse. Spend plans are a weak signal next to the trading calendar they accompany.

LEAD TIME

Thirteen weeks out is barely worse than one week out

Error by weeks ahead, company level



Best method against the benchmark that only knows where demand was last week.

Demand here is driven by things known months in advance: the season, the holiday peak, the flyer schedule. A forecast built only from those does not decay as the horizon lengthens.

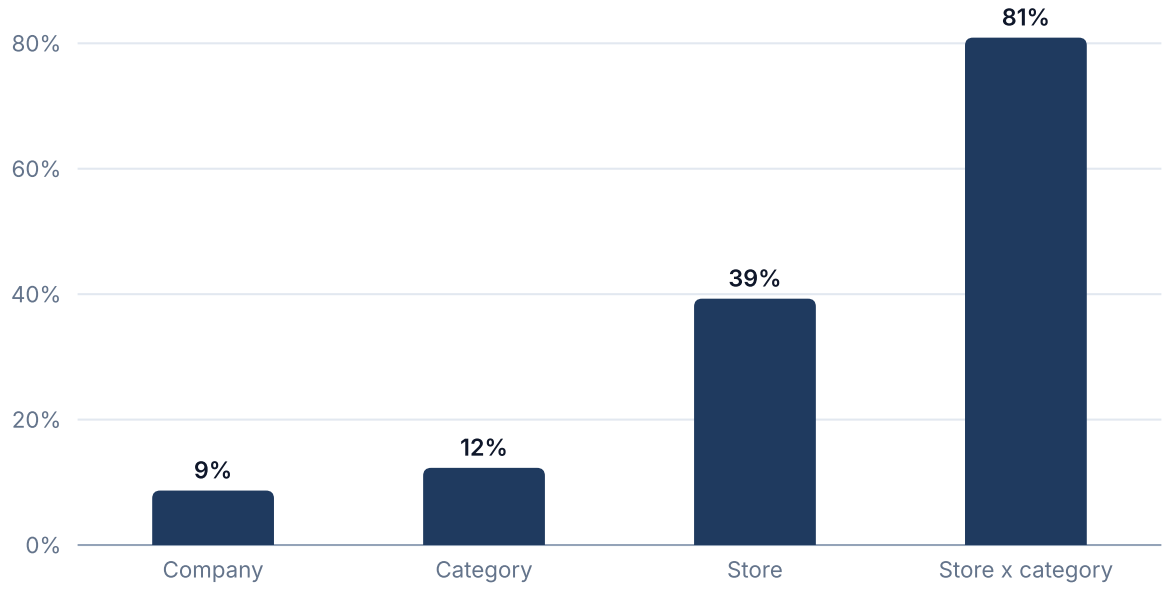
A forecast built from recent demand gains nothing from being made late either: repeating last week is about 5 times worse than the best method at every horizon, one week out and thirteen.

> This is the practical point for planning. Buying decisions made a quarter ahead are not penalised by being made early, so the lead time is a supply-chain constraint, not a forecasting one.

HONEST RANGES

A number with no range cannot be ordered against

Where demand actually lands, as a share of the forecast



The 80% range at each level. Bars show the 10th and 90th percentiles of actual divided by forecast.

LEVEL	80% RANGE	WEEKS INSIDE IT
Company	97% to 106%	93.8%
Category	95% to 107%	80.2%
Store	83% to 122%	79.7%
Store x category	68% to 149%	81.4%

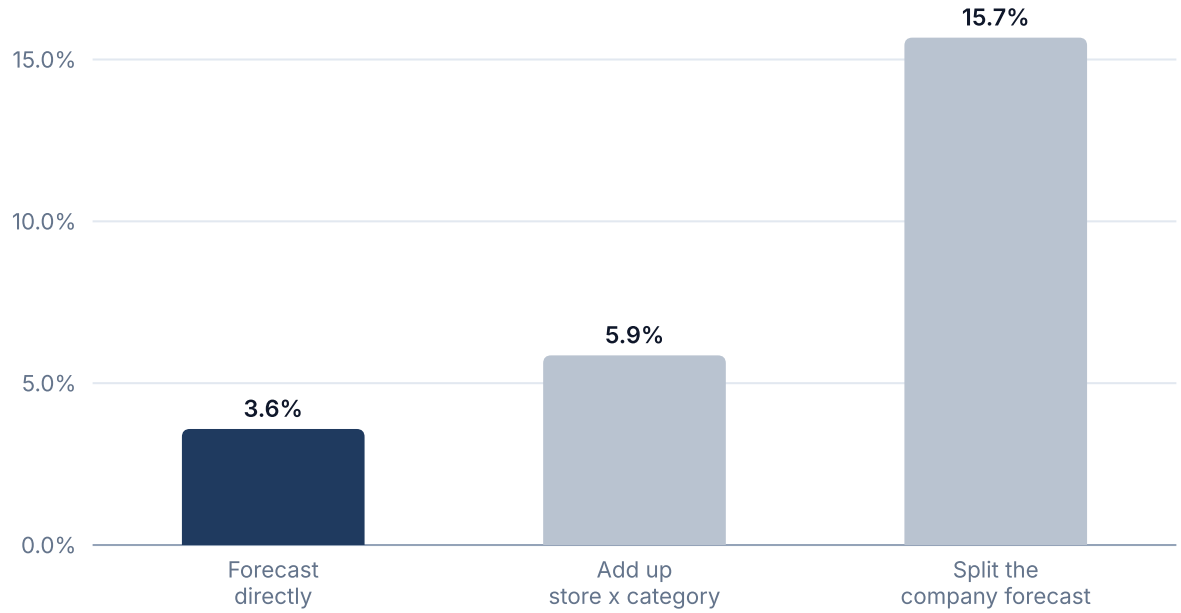
Ranges estimated on the first five forecast origins and tested on the last five, so the check is out of sample.

> Next week's company forecast of $2,961,281 means demand between $2,881,931 and $3,139,096. The same forecast for one category in one store can be out by a third in either direction. Both are usable; only one can be ordered against without a buffer.

HIERARCHY

Forecast each level directly. Do not split and do not stack.

Category-level error, by how the forecast is produced



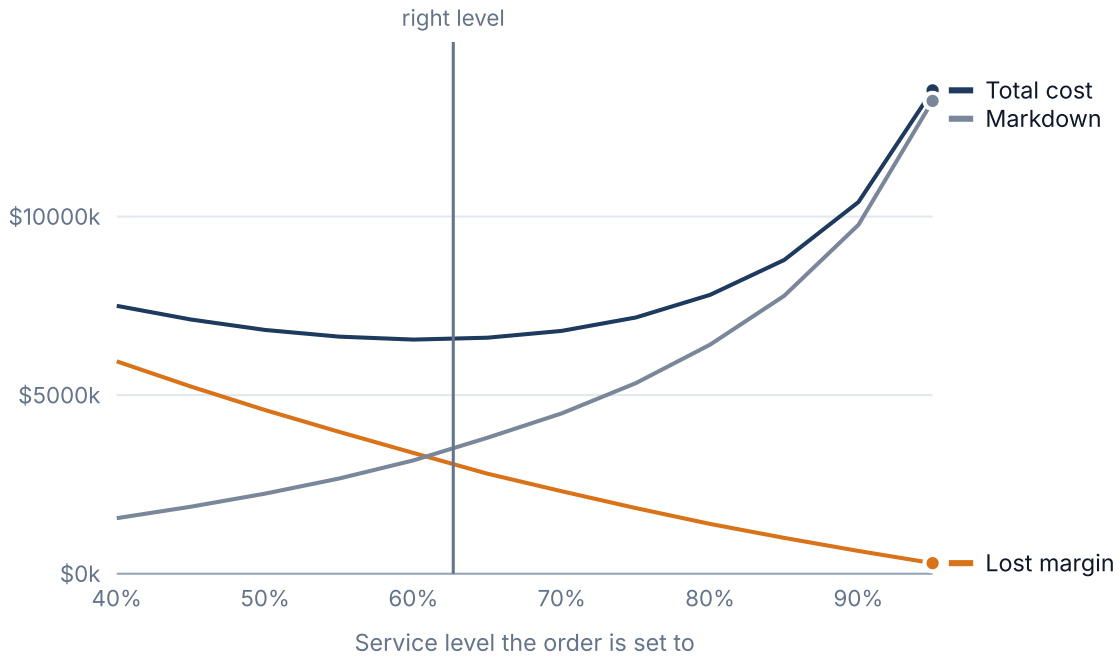
Measured on the five forecast origins held back from fitting the ranges.

- **Splitting the company forecast** by last year's category shares is 4.4 times worse. Categories peak at different times of year, and a single share flattens that.
- **Adding up store-by-category forecasts** is better but still worse than forecasting categories directly: the noise in 480 thin series does not fully cancel.
- **Direct forecasts disagree with each other**, so a reconciliation step is still needed before the numbers are published. Reconcile the published figures; do not let one level generate another.

THE DECISION

Ordering to the forecast is a $256k mistake

Cost of a week's ordering, by service level



Lost margin plus markdown across 480 store-and-category combinations, in CAD. Demand is measured as revenue at full price.

A unit short costs the 42% margin it would have earned. A unit over costs the 25% markdown needed to clear it. Being short hurts more, so the right order is above the forecast, at the 63% point of its range.

ORDERING RULE	COST A YEAR	FILL RATE
Repeat last year	$8793k	87.1%
Order to the forecast	$6825k	87.7%
Order to the service level	**$6569k**	**91.9%**

Total cost of forecast uncertainty is $6.6M a year, or 7.4% of revenue at this level. Halving the remaining error would be worth $3.3M a year, which is the budget any further forecasting work has to earn.

RECOMMENDATIONS

Four actions

1 **Run a method per level, not a model for everything**

Last year adjusted to the current level at company and category; the calendar regression at store and store-by-category. Re-run the comparison every year; the winner can change.

$2.2M a year

2 **Order to the service level, not to the forecast**

Set order quantities at the 63% point of the forecast range. It is a one-line change to the ordering rule and needs no new model.

$256k a year

3 **Publish the range with every number**

A company forecast is good to within a few points; a store and category forecast can be a third out either way. Planners who see only the midpoint will treat the two as equally firm.

Prevents false confidence

4 **Spend the next effort on the trading calendar, not the algorithm**

Knowing the promotion plan earlier and more precisely beat every modelling change tested here. A cleaner events calendar is worth more than a better model.

Up to $3.3M a year

APPENDIX

Method, assumptions and limits

Data. 156 complete Monday weeks, from 6 November 2023 to 1 November 2026, aggregated from every order line. Store-level series count in-store sales only.

Backtest. Ten origins four weeks apart from week 104, each forecasting thirteen weeks. Every model is refitted at every origin on data available at that point.

Methods. Three benchmarks; ridge regression on log demand with trend, five Fourier harmonics, trading-calendar flags and media spend; a boosted model pooled across series; and a simple average of the two best.

Ranges. Empirical quantiles of actual divided by forecast, estimated on the first five origins and tested on the last five.

Assumed, not measured. A 42% gross margin and a 25% markdown to clear overstock. Both are adjustable in the explorer, and the optimal service level moves with them.

Limits. The ordering model treats each week independently: there is no inventory carried between weeks, no lead time and no minimum order quantity. A real replenishment system would soften both the shortfalls and the markdowns, so the cost figures are an upper bound on the benefit of accuracy.

Not tested. Price elasticity and cannibalisation between categories. Demand is forecast as it happened under the actual price and promotion plan, not as a response to a changed one.

Synthetic data. Maple & Main is fictional. Figures illustrate the method, not any real company.

Code. SQL, Python and SAS, with a model card and QC log, are in the project repository.